NOT MEASUREMENT SENSITIVE

MIL-T-46701F
27 March 1990
SUPERSEDING
MIL-T-46701E
5 August 1983

MILITARY SPECIFICATION

TRUCK, TILT-FRAME: CONTAINER HANDLING, WITH TILT CAB, DIESEL ENGINE DRIVEN, 51,000 POUNDS GVW, 6x4, COMMERCIAL; INCLUDING DETACHABLE BODIES AND STATIONARY COMPACTORS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a system consisting of a truck and detachable body(s) for the loading, compacting, storing, transporting, and discharging of refuse and other materials. A subsystem is also covered, consisting of a stationary compactor and a detachable body designed to mate with the stationary compactor. The truck, detachable body(s) and stationary compactor units may be ordered as a system or as individual units to supplement an existing system. The truck covered by this specification is diesel engine driven, six-wheel, four-wheel drive, with a minimum gross vehicle weight (GVW) of 51,000 pounds, mounting a tilt-frame unit and hydraulic system for loading, transporting, and unloading the detachable body. The bodies covered by this specification are specifically designed for use with the truck and tilt-frame unit. In addition, one body is specifically designed for attachment to the stationary compactor to receive refuse and other material during the compaction cycle of the stationary compactor. Items procured under this specification are commercially designed and are warranted by the manufacturer as specified in the acquisition documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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1.2 Classification and equipment. The detachable bodies and stationary compactors shall be of the following types and sizes, as specified (see 6.2):

Type A	- Cargo, open top
Size 8	- 8 cubic yards
Size 15	- 15 cubic yards
Size 20	- 20 cubic yards
Size 30	- 30 cubic yards
Size 40	- 40 cubic yards
Type B	- Cargo, closed top
Size 30	- 30 cubic yards
Size 40	- 40 cubic yards
Type C	- Compaction, short stroke (not less than 33 cubic yards)
Type E	- Self loading and compaction (not less than 30 cubic yards)
Type F	- Cargo, closed top, for use with stationary compactors (not less than 42.5 cubic yards)
Type G	- Stationary compactor
Size 1.5	- 1.5 cubic yards
Size 4	- 4 cubic yards

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
FEDERAL

A-A-393	- Extinguisher, Fire, Dry-Chemical (Hand-Portable).
W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting, and Starting.
W-F-800	- Fuel Oil, Diesel.

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	MIL-T-704	- Treatment and Painting of Material.
	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
	MIL-R-23954	- Refuse Containers, Hoisting Unit; Tilt-Over, For Front Loading Compaction Type Trucks.
	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
*	MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, Metric, NATO Code Number H537.

STANDARDS

FEDERAL

	FED-STD-595	- Colors.

MILITARY

	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.

* (Unless otherwise specified, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Form Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for a hard copy, or microfiche copies are available from the Director, Navy Publications and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

* (Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issues of the non-Government documents cited in the solicitation (see 6.2).

* AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Z245.1 - Safety Requirements for Refuse Collecting and Compacting Equipment.
* Z245.3 - Safety Requirements for Stability of Refuse Bins.

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code, Section VIII.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM STANDARDS
B-117 - Method of Salt Spray (Fog) Testing.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

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North Atlantic Treaty Organization (NATO)
Allied Engineering Publication (AEP-5)
Part II - NATO Standard Diesel and Spark Ignition Engines Laboratory Test, dated January 1984.

(Application for copies of NATO publications should be addressed to North Atlantic Treaty Organization, Military Agency for Standardization, 1110 Brussels, Belgium.)"

NATIONAL SOLID WASTES MANAGEMENT ASSOCIATION (NSWMA)
Listing of NSWMA-Rated Commercial/Industrial Stationary Compactors.

(Application for copies of NSWMA publications should be addressed to NSWMA, 1120 Connecticut Ave., NW, Washington, D.C. 20036.)

SAE, INC.
SAE Standards and Recommended Practices

	J537	- Storage Batteries.
*	J551	- Performance Levels and and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
	J688	- Truck Ability Prediction Procedure (DoD adopted).
	J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs.
	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book

(Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the document. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.6, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.6, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies, mounted equipment, and bodies shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models, components, and other equipment and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of the invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle, detachable bodies, stationary compactor and components specified in 3.1, special equipment shall be furnished as specified herein.

3.1.1.1 Painting and marking. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, and data plates shall be in accordance with MIL-STD-1223, except for the following:

(a) The exterior of the tilt frame and associated equipment and the exterior of the detachable body and stationary compactor shall be treated and painted in accordance with type A of MIL-T-704
(b) The interior of the detachable body and the stationary compactor shall be painted with rust inhibiting primer, in accordance with the manufacturer's commercial practice
(c) The color of the exterior of the vehicle and the tilt-frame and associated equipment, and the exterior as well as the interior of the detachable body and stationary compactor, shall match color chip No. 13538, gloss yellow of FED-STD-595 for the Navy; color chip No. 16081, gloss gray, of FED-STD-595 for the Air Force; and color chip No. 17875, gloss white of FED-STD-595 for all other services
(d) Vehicle identification markings shall be applied to the tilt-frame truck, and shall not be applied to the detachable bodies
(e) The color of vehicle identification markings shall be gloss black No. 17038 for the Navy; reflective white numerals and letters on a red background for the Air Force; and gloss black No. 17038 numerals and letters on a reflective white background on vehicle and on front and rear marking plates for all other services
(f) Rubber seals and hydraulic hoses need not be painted.

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* 3.1.1.2 Rustproofing. When specified (see 6.2), the truck cab and chassis shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished for the truck cab and chassis.

3.1.1.3 Drain plugs. Drain plugs installed in the rear axle shall be of the permanent magnet type.

* 3.1.1.4 Towing devices. Not less than two hooks, loops, pins, or a single center-mounted towing eye for towing the vehicle shall be furnished on the front of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.5 Splash and stone throw protection. Rear wheel splash and stone throw protection in accordance with SAE J682 shall be furnished. Rubber mud flaps shall be provided to the rear of the front wheels. A metal strip not less than 1/8 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flaps. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1-1/4 inch wide by 3/32 inch thick shall be furnished at each bolt.

* 3.1.1.6 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound level shall not exceed 84db(A). The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

* 3.2.4 Safety standards. The vehicle, mounted equipment and stationary packers shall comply with all OSHA requirements and with ANSI Z245.1 and Z245.3.

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3.2.5 Curb weight. The curb weight shall include the weight of the chassis and cab with all attachments; accessories; equipment; the tilt-frame unit, excluding the detachable body; and a full complement of fuel, lubricants, and coolant.

3.2.6 Gross vehicle weight. The GVW shall consist of the curb weight, operator weight (computed at 175 pounds) and gross payload in transporting position to provide not less than the specified GVW (see 3.2.9).

3.2.7 Gross payload. The gross payload shall be the combined weight of the detachable body and the payload, defined as the weight of refuse or other material evenly distributed in the body load area.

3.2.8 Component capacities. The total vehicle suspension, axle and tire rated capacities, (gross axle weight ratings) shall be not less than 18,000 pounds on the front and 44,000 pounds on the rear.

3.2.9 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component, vehicle, and tilt-frame unit ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the engineering office of the procuring activity. The minimum GVW rating shall be 51,000 pounds.

3.2.10 Cab-to-trunnion dimension. The cab-to-trunnion dimension shall be in accordance with the recommendations of the manufacturer of the tilt-frame unit and the detachable bodies.

3.2.11 Overall width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors, and tires, shall be not more than 96 inches. The width over-the-tires shall be not more than 100 inches.

* 3.2.12 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.13 Asbestos. Asbestos materials shall not be used in any form in any part of the vehicle.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the truck loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend a 0.50 percent grade at 50 miles per hour (mph). Gradeability requirements shall be met with the transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

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3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of the manufacturer's recommended governed speed, and shall provide a vehicle speed of not more than 3 mph.

* 3.3.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed shall be determined in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52, under all conditions of loading, up to specified GVW.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. The vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine with not less than six cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable. A test report shall be available showing that a representative engine has been tested and certified to meet the requirements of NATO AEP-5 test procedures (400 hours dynamometer test to assure JP-8 compatability.)

3.4.1.2 Oil filter. A full-flow type oil filter shall be furnished.

3.4.1.3 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.4 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity.

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* 3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. Control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.6 Fan clutch. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be provided. The fan clutch shall be asbestos free.

3.4.1.7 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.8 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater; an engine oil heater; and a fuel warmer shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110 volts alternating current (ac), and shall be wired through a junction block, including a fuse or a circuit breaker, to a single three-pronged (male), weatherproof slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heaters simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F), without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose, excess wire. A carrier for the connecting cable shall be mounted within the cab or engine compartment and shall provide positive cable retention during vehicle operation. Heaters shall be furnished as follows:

(a) A coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. Engine thermostat with an operating range of 170°F to 195°F shall be installed
(b) The immersion type engine oil heater 300-watt minimum rating, with 170°F to 195°F thermostat, shall be installed in oil pan through any convenient opening
(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F
(d) A fuel warmer or pre-heater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than

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the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. A 12- or 24-volt direct current (dc) starting system with a 12-volt lighting system shall be furnished. A minimum 60 ampere alternator shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.2 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and total cold cranking ampere rating at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table I. Batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE I. Batteries.

	Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
	Diesel (less than 220 gross horsepower)	320	1,200
*	Diesel (220 gross horsepower and over)	480	1,740

* 3.4.2.3 Lighting. The electrical system shall be 12 volts. All vehicle lights and reflectors shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on rub rails (unless recessed and fully protected) nor on vehicle bumpers.

* 3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589 class A and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have a visible and audible type flash indicator.

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3.4.2.5 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. The engine shall be equipped with an air cleaner. When specified (see 6.2), a single stage, or two-stage, dry type air cleaner shall be furnished with a dash mounted indicator to determine when the filter should be changed.

3.4.3.2 Fuel tanks. Fuel tanks shall be of not less than 50 gallon total capacity (43 for vehicles destined for California). When more than one tank is furnished, means shall be provided to assure an equalized fuel level in both tanks.

* 3.4.3.3 Fuel and water separator. The manufacturer's standard or optional fuel filter shall be provided. When specified (see 6.2), a fuel and water separator shall be furnished. The separator shall include a water coalescer and a drain valve. When a separator is required, a combination filter/separator unit may be furnished.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust system shall be of the vertical type. Vertical mufflers capable of being reached easily by personnel entering or leaving either side of the cab shall be provided with a heat shield. Vertical exhaust pipes shall be furnished with a hinged rain cap.

3.4.5 Automatic or semiautomatic transmission. The vehicle shall be provided with an automatic or semiautomatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The automatic or semiautomatic transmission furnished shall be of the fixed ratio type and shall include a hydraulic torque converter and not less than five forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with an SAE J704 power takeoff opening.

* 3.4.5.1 Power takeoff. A power takeoff shall be provided. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

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3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the vehicle furnished. The vehicle shall have a frame with reinforcement extending at least from the centerline of the front axle to the rear of the frame. Reinforcement shall provide sufficient structural strength in the chassis frame, through increased resisting bending moment, to at least equal the loads imposed with the vehicle loaded to specified GVW.

3.4.8 Suspension. The vehicle shall be equipped with suspension system components having a rated capacity at least equal to that specified in 3.2.8. When spring capacity is rated at the spring pads, unsprung weight shall be deducted.

* 3.4.9 Axles. Axle ratings shall be at least equal to that specified in 3.2.8. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Rear bogie. Unless otherwise specified, the rear bogie shall be of the four-wheel type complete with axles, springs, torque rods and all other necessary parts. When specified (see 6.2), a walking beam type suspension with rubber load cushions shall be furnished in lieu of the spring type. The bogie shall be provided with means permitting differential action between the two axles, and a manually controlled lockout assuring equal power to each axle. Lockout control shall be located in the truck cab. Axle gear ratios, in combination with transmission ratios, shall provide performance specified in 3.3.1 through 3.3.1.3.

3.4.9.2 Traction control. Traction control shall be furnished on one rear axle. The traction control shall actuate automatically to ensure that power is transmitted to the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished on front axles rated at 14,000 pounds or less or on rear axles. When specified (see 6.2), 10-stud disc-type wheels shall be furnished.

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* 3.4.10.1 Tires. Steel belted radial-ply tires or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall have highway tread. Tires shall be of the tubeless type or, on axles where multi-piece rims are permitted (see 3.4.10), may be of the tube type. Tires shall be of rated capacity at least equal to that specified in 3.2.8. Tires and tire size designation systems shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.4.10.2 Inner tubes. When tube type tires are permitted herein or specified in procurement documents, inner tubes shall be of heavy duty type and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b) and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

* 3.4.11.1 Air brakes. The vehicle shall be equipped with full air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute (cfm)
(b) Air storage reservoir(s), each tank equipped with a drain, and with safety and check valves between the compressor and last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low-air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
* (i) Automatic slack adjusters on cam type brakes or internal self-adjusting slack adjusters on wedge and disc type brakes.

* 3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete

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with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), and additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.11.3 Increased braking capability. When increased braking capability is specified (see 6.2), the service brakes shall be augmented by one of the following braking systems:

(a) an electromagnetic driveshaft retarder
(b) a system which opens all or some of the engine exhaust values near the end of the compression stroke, thereby converting vehicle motion to a pumping loss
(c) a controlled gate valve in the exhaust manifold, which produces back pressure on the engine pistons during the exhaust stroke
* (d) a hydrodynamic retarder integral with the automatic transmission.

A dash mounted switch shall be provided to activate, modulate, or cut out the brake augmentation. The switch shall be marked to indicate its position. When active, the system shall be fully controlled by means of the conventional driving controls to apply retardation during vehicle deceleration, and to cut it out in the other operating modes. Any of the three systems shall permit the vehicle to negotiate a 6 percent down grade without exceeding 27 miles per hour, within the operating restrictions of the engine and transmission and without the use of service brakes. For (b) or (c), above, the retarder shall be approved by the engine manufacturer.

* 3.4.12 Cab. The manufacturer's standard forward tilting cab shall be furnished complete with forward tilting and locking mechanisms. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. The cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with at least one door equipped with an external, key operated lock. Drip rails shall be provided above the cab doors. Grab handles shall be provided on each side of the cab to assist personnel in entering and leaving the cab. Provisions for cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield. When the front tires extend beyond the cab fenders, rubber fender extensions extending to at least the outside of the tire tread shall be furnished.

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* 3.4.12.1 Cab interior. Interior lighting shall be provided. Unless otherwise specified, the cab shall have an upholstered, full-width, adjustable seat and back or an individual, adjustable, driver's seat and an individual passenger seat. When specified (see 6.2), an individual adjustable driver's seat and an individual passenger seat shall be furnished. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Seat belts for each seating space shall be installed.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by air or electric motor(s).

3.4.15 Bumper. A channel type front bumper, or a front bumper integral with the cab, shall be provided.

3.4.16 Tool stowage. Stowage space of sufficient size to accommodate hand tools, antiskid chains, and emergency reflective triangles shall be furnished for stowage of equipment during vehicle operation. When stowage space is located outside of the cab, it shall be weatherproof and shall provide for padlocking.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms inside of cab shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Openings in the cab, for the purpose of installing control levers, shall be sealed with a boot or other flexible type cover to prevent roadwash and dirt from entering the cab. All instruments inside of the cab shall be visible to the driver when seated in the driving position.

* 3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key operated ignition switch
(b) Ammeter, voltmeter or charging indicator
(c) Fuel gage
* (d) Oil pressure gage or red indicator warning light

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* (e) Engine temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Tachometer
(i) Ash receptacle
(j) Dual sunvisors
(k) Driver's compartment ventilator other than window.

3.4.20 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air-operated horn shall be furnished.

3.4.21 Engine hour meter. An engine hour meter, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to assure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

* 3.4.22 Rearview mirrors. Outside rearview mirrors, each with at least two supports, shall be mounted on each side of the cab. Combination (flat and convex) mirrors, enclosed in a common housing, shall be furnished. Each combination mirror assembly shall have at least 50 square inches of flat reflective area and a convex surface having at least 25 square inches of reflective area.

3.4.23 Back-up alarm. When specified (see 6.2), an audible, pulsating, signaling device (mechanical or electrical) shall be provided to caution personnel when the vehicle is in reverse gear operation.

3.4.24 Fire extinguisher. A 5-pound dry chemical type fire extinguisher, conforming to Federal Motor Carrier Safety Regulation 393.95(a), shall be mounted convenient to the driver. The fire extinguisher shall have an Underwriter's Laboratory rating of 2A:10B:C or more, shall include a metal safety pin and wire and shall conform to type I, class 1, size 5 of A-A-393. The fire extinguisher shall permit visual determination as to whether it is fully charged.

3.4.25 Periscopic mirror screen. A periscope type mirror screen shall be installed in the ceiling of the cab to enable the operator to view the containers and detachable bodies during all stages of body loading and container dumping operations.

3.5 Tilt-frame unit. The tilt-frame unit, mounted on the truck chassis, shall be capable of handling types A, B, C, and F and all the sizes of bodies classified in 1.2. When specified (see 6.2), the tilt-frame unit shall also

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be capable of handling type E, self-loading and compaction type bodies. When specified (see 3.5.1.2.1), the tilt-frame unit shall also be capable of handling tandem mounted type A and B cargo bodies each not exceeding 126 inches in length. All operations of the unit, that is, pulling up the detachable body; placing it in the carrying position; transporting it to the destination; dumping the contents of bodies with body still mounted on the unit; placing it back on the ground (loaded or empty); or when specified (see 6.2), placing on a dock not less than 48 inches high; shall be capable of being controlled by one man with controls within the truck cab. The operator shall not require any assistance nor be required to leave the truck cab except to release or fasten body rear door lock and, for cable operated unit, to fasten or unfasten the chain or cable from the body.

3.5.1 Design. The tilt-frame unit shall consist of a tilt-frame with rollers or replaceable wear strips. The unit shall be so designed and balanced on the truck chassis as to operate satisfactorily without truck counterweights. The unit shall be provided with automatic locks for securing the detachable bodies while in transit. Controls shall be in the truck cab.

3.5.1.1 Tilt-frame. The tilt-frame shall be hinged at the rear. The front of the tilt frame shall be raised and lowered by means of double-acting hydraulic cylinders of suitable capacity. The cylinders shall be capable of lifting the tilt-frame and loaded body to an angle of not less than 45 degrees from the top of the chassis frame.

3.5.1.2 Body control mechanism. Unless otherwise specified (see 6.2), movement of the detachable body during pull-up and set-down operations shall be controlled by double-acting hydraulic cylinders, by hydraulically operated winch and cables, or by hydraulic cylinder operated sheave and cable system. The body movement control assembly shall maintain positive control of the body at all times during the pull-up and set-down operations. When a winch system is used, the winch shall be equipped with a cable tensioner.

3.5.1.2.1 Tandem loading device. When specified (see 6.2), a latch or coupling device designed to facilitate tandem loading of two (2) short detachable bodies shall be furnished.

3.5.1.2.2 Retaining device. When a hydraulic cylinder loading system is furnished, or when specified (see 6.2), a secondary mechanical device designed to prevent movement of the detachable body, if disengaged during pull-up and set-down operations, shall be provided.

3.5.1.3 Stabilizing jack(s). When required to prevent the front wheels from leaving the ground with the rated payload, hydraulically operated, roller type stabilizing jack(s) shall be installed at the rear of the mechanism to assist in stabilizing the truck during loading, dumping, and unloading operations.

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3.5.1.4 Fenders. The tilt-frame unit manufacturer's standard fenders shall be provided over the rear tires. The fenders shall have sufficient clearance to allow operation of the truck with antiskid chains on the rear tires.

3.5.2 Capacity. The minimum rated pickup capacity of the tilt-frame unit shall be 40,000 pounds.

3.5.3 Pickup height. The tilt frame, mounted on the chassis, and with all chassis wheels at ground level, shall be capable of picking up, and when specified (see 6.2), setting down, detachable bodies from any point between ground level and a loading dock not less than 48 inches high.

* 3.5.4 Hydraulic system. Power shall be transmitted from a transmission mounted power takeoff, through a heavy duty gear or vane-type pump. Capacity and pressure ratings shall be sufficient to accomplish loading, unloading, or discharging operations within a period of two minutes. Relief valves shall be incorporated in the hydraulic system to prevent excessive pressure. A line tap shall be provided so that relief pressure may be checked periodically. All hydraulic lines, hoses, and fittings shall meet or exceed the requirements of SAE recommended practices and standards. The exhaust system shall be shielded so that, in the event of a hydraulic line rupture, hydraulic fluid will not come into contact with any part of the vehicle exhaust system. The size, volumetric capacity and pressure ratings of all components shall be such as to assure smooth and efficient operation of the system. All surfaces of the hydraulic ram piston rods shall be chrome plated full length with sufficient thickness to withstand a 30 hour salt spray test in accordance with ASTM B-117. The connecting ends shall be treated in accordance with the best commercial practice.

* 3.5.4.1 Hydraulic fluid. Unless otherwise specified, the manufacturer's standard or recommended hydraulic fluid shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system. A permanent decal located near the hydraulic system fill port shall be furnished stating "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-83282 ONLY."

3.5.4.2 Intersystem connections. A complete set of cab controls and all equipment necessary for connecting the basic hydraulic system to the hydraulic system of the type C bodies (see 3.6) shall be provided. All connections shall be of the quick-coupling type and shall be permanently coded to minimize the possibility of an incorrect connection. When specified (see 3.5), the cab controls and intersystem connections shall also be provided for type E bodies.

3.6 Detachable bodies and stationary compactors. Detachable bodies or stationary compactors of the type(s) and size(s) as specified (see 1.2) shall be furnished. Bodies shall be functionally compatible with and suitable for

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use with the truck and tilt-frame unit specified herein. Type F detachable bodies shall be functionally compatible with and suitable for use with the stationary compactors specified herein. Stationary compactors shall be functionally compatible with and suitable for use with the type F bodies specified herein. Functional compatibility and suitability for use shall include full inter-operations of the truck, the tilt frame, the detachable bodies and the stationary compactors as specified herein without modification or alteration of any part of the system by either the Government or the contractor. When specified (see 6.2), to supplement an existing system, functional compatibility and suitability for use as defined herein shall be extended to include those parts of the existing system specified by the procuring activity.

3.6.1 Types A and B (cargo). Type A and B bodies shall be box shaped with side-hinged rear door(s). Net cubic yard capacity shall be not less than that required for the size body furnished (see 1.2). Type A body shall have an open top. Type B body shall have a closed top with counterbalanced lids, and shall provide a single opening of at least 70 percent of the horizontal cross-sectional area of the body. Type B top lids shall be hinged on the streetside of the body and each lid shall be capable of being individually opened without interfering with adjacent lids. Type B body shall have a top walkway to allow the operator to readily open the top lids. The maximum width of each body shall be 96 inches. The height of the body shall be such that, with the body empty and resting on the tilt-frame truck specified above, the overall height measured from the ground to the top of the body shall not exceed 13 feet 6 inches. Overall length shall be within the body length limits recommended for use with the tilt-frame truck.

3.6.1.1 Design. Each body shall have an understructure that will allow it to be picked up and pulled onto the truck without requiring any undue jerking of the truck. The design shall permit, after release of the end gate lock, complete discharge of contents by the operator using controls located in the truck cab, without assistance, while the body is mounted on the truck. There shall be no constriction or obstruction throughout the interior of the body to retard or interfere with the dumping of the contents in the discharge action.

3.6.1.2 Construction. The sides, ends, and bottom of each body shall be sufficiently stiffened and reinforced to withstand the stresses of capacity loading under normal usage without permanent set. The body shall have an understructure suitably constructed of properly reinforced members to handle capacity loads under normal usage without distortion and undue stress. All interior surfaces shall be smooth and easily cleanable. The body shall be constructed of not less than 10 gage (0.1345 inch) sheet steel throughout. For type A body, a header shall be furnished at the rear of the body to prevent spreading of body sides. The header shall be removable and shall not interfere with the operation of the rear door. When specified (see 6.2), the entire body shall be liquid-tight to a height of at least 14 inches.

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3.6.1.3 Discharge openings. Discharge of the contents of the body shall be through the rear (the end of the body at the rear of the truck when in normal transporting position). The body shall have one or two side-hinged rear doors. Provisions shall be made for keeping the discharge door or doors tightly closed while the body is located at the pick-up station and in transit. Each door shall be properly reinforced and shall be attached to the body with the equivalent of at least three heavy strap hinges. The effective discharge opening area for the body shall be not less than 95 percent of the largest vertical (when in normal position) cross-sectional area of the body. Type A body shall be furnished with an escape ladder permanently mounted inside the body.

3.6.1.4 Accessories. When specified (see 6.2), the body shall be fitted with four crane eyelets of sufficient strength to withstand lifting of a fully loaded body. Type A bodies shall be provided with suitable means for securing a tarpaulin used for covering the exposed top of the loaded body. When specified (see 6.2), a tarpaulin shall be furnished. The tarpaulin shall be of such size as to cover the top of the body completely. Suitable means shall be provided for attaching the edges of the tarpaulin to the outside walls of the body.

3.6.1.5 Reflectors. The body shall have reflectors mounted in accordance with Federal Motor Carrier Safety Regulations 393.12 and 393.26(a) through (d).

3.6.2 Type C (compaction, short stroke). Type C detachable body shall be a compaction type with the compaction plate traveling the length of the compression chamber.

3.6.2.1 Design. The compaction body shall consist of a compression chamber and storage chamber, the combined volume of which shall be not less than 33 cubic yards. The compression chamber shall be loaded from the top or sides and the refuse rammed into the storage chamber. The compaction cycle shall be automatic, manually initiated for each cycle. The packing mechanism shall provide a packing force of not less than 29,000 pounds. The maximum overall dimensions of the body shall be as specified in table II:

TABLE II. Type C body dimensions.

Designation	Dimension
Length	294 inches
Height	97 inches
Width	96 inches
Side loading height	45-1/2 inches

3.6.2.2 Construction. The body shall be constructed on an understructure which permits pickup and unloading operations to be performed by the tilt-frame unit specified herein (see 3.5 through 3.5.4.2). The body

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shall be constructed of not less than 10 gage (0.1345 inch) sheet steel, reinforced for strength. The body shall be reinforced by box braces, vertically and across the top of the storage area. The packer assembly shall be fabricated of not less than 10 gage (0.1345 inch) sheet steel, properly reinforced by heavy footing angles and flanged gussets. The floor shall be constructed of 3/16-inch steel supported by 10 gage high tensile steel pressed crossmembers to withstand the impact of top loading.

3.6.2.3 Load openings. The body shall be provided with a load opening of approximately 75 percent of the total compression chamber area. The opening shall be covered with lids fabricated of not less than 16 gage (0.0598 inch) sheet steel reinforced and cross-braced to form a firm lid of adequate strength for the intended service. The lids shall be hinged to form one integral unit.

3.6.2.4 Discharge openings. The discharge of contents from the body shall be accomplished by tilting the frame upward. The body shall have one or two side hinged rear doors. Provisions shall be made for keeping the discharge door(s) tightly closed while the body is located at the pick-up station and in transit. Each door shall be properly reinforced and shall be attached to the body with the equivalent of at least three heavy strap hinges. The effective discharge opening area for the body shall be not less than 95 percent of the largest vertical (when in normal position) cross-sectional area of the body.

3.6.2.5 Power unit. The compaction body shall be equipped with an electrically powered independent hydraulic system including pump, oil reservoir, and compaction cylinder. Unless otherwise specified (see 6.2), power input shall be 208 volts, ac, 3-phase, 60 hertz (Hz).

3.6.2.6 Reflectors. The body shall have reflectors mounted in accordance with Federal Motor Carrier Safety Regulations 393.12 and 393.26(a) through (d).

3.6.3 Type E (self-loading). Type E detachable body shall be a compaction type with front hoisting arms, and with the compaction and ejection plate traveling the full length of the body.

3.6.3.1 Packer body. The compaction body shall be constructed on an understructure which permits pick-up and unloading operations to be performed by the tilt-frame unit specified herein (see 3.5 through 3.5.4.2). When subjected to the operating and loading conditions specified, the load carrying members of the understructure and body shall exhibit the capacity to resist the imposed bending, twisting, and buckling forces without permanent set or part failures. The body top and side walls shall be constructed of not less than 12 gage (0.1046 inch) high-tensile sheet steel. The body floor shall be fabricated of not less than 1/4 inch high-tensile sheet steel. The body top and side walls shall be reinforced with box braces, and the floor

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shall be reinforced with structural steel members. The span of unsupported floor between underfloor crossmembers shall be not more than 12 inches. The body inside top shall have at least two rows of retainer teeth to prevent refuse from falling forward after the compaction stroke. The interior surfaces of the body shall be smooth to expedite ejection and cleaning operations. The entire body shall be liquid-tight to a height of at least 5 inches. The rear (discharge) opening shall be equivalent to the inside dimensions, reinforced around the edges, and provided with a top-hinged door. The loading hopper opening shall be not less than 80 inches by 80 inches, framed around the edges to resist denting during loading operations, and furnished with a hinged cover. The curbside opening shall be not less than 30 inches by 30 inches, reinforced around the edges. The opening shall be covered by a sliding door held in closed position by a lever type, spring loaded latch. A means shall be provided to prevent movement of the packer plate when the curbside (charging) door is open. The body shall withstand the maximum pressure exerted by the packer plate upon the refuse without permanent deformation.

3.6.3.1.1 Volume and dimensions. The capacity of the body shall be not less than 30 cubic yards. Maximum overall dimensions shall be as specified in table III:

TABLE III. Type E body dimensions.

Designation	Dimension
Length	270 inches
Width	96 inches
Height	96 inches

3.6.3.1.2 Hopper cover. The hopper cover shall be operated (opened and closed) by a hydraulic cylinder with control in the cab. The hopper cover shall be reinforced. A warning light shall be furnished in the cab to indicate to the operator that the hopper cover is in the open position.

3.6.3.1.3 Wind guard. Metal hopper extensions to act as a wind guard shall be provided at hopper sides and rear to minimize the possibility of spillage that may occur on windy days when containers are being dumped. The wind guard shall be at least the full length of the hopper opening and shall extend a minimum of 24 inches above the body. The wind guard shall provide not less than a 270-degree shield around the rear and sides of the hopper opening and shall provide not less than 90 square feet of protection.

3.6.3.1.4 Discharge door. The rear discharge door shall offer an unrestricted opening, shall be reinforced to prevent deformation which may cause leaks between the door and body, shall be fitted with heavy duty hinges at the top, and shall be equipped with a heavy duty cam type lock operable by a single lever.

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3.6.3.2 Compaction and ejection plate. The compaction plate shall permit smooth operation without bypassing refuse during operation, and shall ensure the packing of refuse in all body areas to the rear of the plate. The packer plate assembly shall be powered for compaction, unloading and retraction by a hydraulic telescopic-action cylinder exerting at the end of its first stage not less than 55,000 pounds of horizontal thrust with the body in a horizontal position. The ejection plate shall travel the full length of the body. The system shall permit the hoisting mechanism (see 3.6.3.3) to dump refuse only when the packer plate is in the fully retracted position. When in the fully retracted position, the upper edge of the packer plate shall be forward of the front edge of the hopper opening. The compaction and ejection plate shall completely unload the body without outside assistance. A warning light shall be furnished in the cab to indicate that the packer is in its compaction cycle.

3.6.3.3 Container hoisting mechanism. The lifting and dumping action of the body unit hoisting mechanism shall be accomplished by a double-acting hydraulic cylinder. The capacity of the hoisting mechanism shall be not less than 4,500 pounds. Hoist capacity rating shall be based on the gross load located 30 inches forward of the position of the rear container wall with the container in the pick-up position at ground level and the detachable body mounted on the tilt-frame truck. The unit shall be capable of engaging, lifting, dumping, returning, and disengaging the containers from the ground and from loading docks not less than 48 inches above ground level. The hoisting mechanism shall be capable of handling containers of any size and type of applicable class as specified in MIL-R-23954 without modification of the containers. When specified (see 6.2), to supplement existing refuse collection system, the hoisting unit shall be compatible with the existing system and capable of handling a specific class of container specified in MIL-R-23954 without modification of the container.

3.6.3.3.1 Controls and safety. The hoisting mechanism controls shall be located in the cab of the tilt-frame truck and shall be of the deadman type. The position of the container and hoist arms shall be under the control of the operator at all times, so that the operator can start, stop, or reverse the movement at any point in the cycle. A safety feature shall be designed into the hoisting system to minimize the possibility of injury to operating personnel from the hoist arms, and shall be designed to allow safe entrance and exit to and from the cab by the operator and passenger(s) during any phase of the hoisting operation.

3.6.3.4 Sump. A liquid-collection tank (sump) of not less than 20 gallons capacity, and equipped with a sealed lever-operated hinged gate or a large faucet, shall be provided to receive drainage from wet refuse.

* 3.6.3.5 Hydraulic system. The container-hoisting, compaction and ejection hydraulic system shall be compatible for use with the hydraulic system of the tilt-frame unit specified in 3.5. A complete set of intersystem connections and hydraulic hoses shall be provided. All surfaces

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of hydraulic piston rods shall be chrome plated full length with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM B-117. The connecting ends shall be treated in accordance with the best commercial practice.

3.6.3.6 Wash-out system. A reservoir of not less than 80 gallon capacity shall be installed for washing and disinfecting the interior of the body and refuse containers. Reservoirs shall include a fill valve or opening with filler neck and a discharge outlet. The fill opening shall be covered with a cap that has a handle or T-bar to enable closing and opening by hand. When the reservoir furnished is the pressurized type, a pressure-type cap shall be supplied, and a caution plate shall be provided to warn that pressure must be relieved before opening the cap. Washing solution shall be discharged under pressure from either the compressed air or equivalent type mechanical pump. When compressed air is utilized to provide discharge pressure, the reservoir shall be pressure-vessel designed for an operating pressure of not less than 100 pounds per square inch gage. The air system shall be furnished with a safety relief valve, a dial pressure gage, an air charging valve, and a bleed valve. The bleed valve shall be capable of bleeding off the system pressure in not more than five minutes. The air operated reservoir shall be connected to the truck chassis compressed air system and shall incorporate an air shut-off valve between the compressed air and the liquid reservoir. The mechanical system shall include a reservoir vent to ensure unrestricted discharge of washing solution. Pressure vessels shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code Section VIII for Unfired Pressurized Vessels. Pressure vessels shall be ASME inspected, stamped and registered with the Board of Pressure Vessel Inspection. The tank discharge outlet shall be provided with 50 feet of rubber hose fitted with a spring-actuated shut-off spray nozzle. The hose shall be capable of withstanding the wash-out system pressure. A hose rack shall be provided on the body for storage and carrying of the hose.

3.6.3.7 Reflectors. The body shall have reflectors in accordance with Federal Motor Carrier Safety Regulations 393.12 and 393.26(a) through (d).

3.6.4 Type F (for use with stationary compactor). Type F detachable body shall be of the reinforced compaction type, designed for use with the type G stationary compactor specified herein or, when specified (see 6.2), with the stationary packer specified. The body shall be capable of being detached from the stationary packer, transported and discharged by using the tilt frame of the truck and returned and reconnected to the stationary packer. The detachable body shall be of a tapered-rectangular shape, with greater width at the compactor mating end than at the cab end. The body capacity shall be not less than 42.5 cubic yards. The maximum overall body dimensions shall be as specified in table IV:

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TABLE IV. Type F body dimensions.

Designation	Dimension
Length	278 inches
Width	96 inches
Height	13 feet 6 inches, measured from ground to top of body when on the tilt-frame truck in traveling position with the body empty.

3.6.4.1 Body mounting. The body shall be mounted on a heavy duty understructure which shall support bridge-action of the body during pick-up and set-down operations. For mating both truck and compactor, the body shall be compatible with the truck, the tilt frame, and the container handling and stationary packer guide tracks as specified herein.

3.6.4.2 Construction. The detachable body shall be of reinforced all-steel construction. The walls shall be reinforced by vertical channel or continuously welded box brace members spaced on a maximum of 24-inch centers. The vertical channels or box braces shall meet equivalent members across the top to form continuous bracing from side to side. Additional bracing shall be provided for stresses applied in the areas of the stationary compactor locks. Means shall be provided for engaging the locking mechanism on the stationary compactor (see 3.6.5.5). Additional means shall be provided for engaging the retaining device on the tilt-frame truck (see 3.5.1.2.2). The floor shall be braced with a minimum of 3-inch structural channel crossmembers. Heavy duty longitudinal members shall be provided to support the body from front to rear. The longitudinal members shall be of sufficient strength, number and spacing to resist bridging action and to provide proper floor support. All interior surfaces shall be smooth and free of obstructions which might restrict discharge of the load and shall be easily cleanable.

3.6.4.3 Discharge. Discharge of the compacted load shall be through the rear (the end of the body at the rear of the truck when in normal transporting position), by tilting the frame upward. The tapered width shall ensure thorough discharge of the compacted load when the body is raised to the tilt dump position.

3.6.4.3.1 Rear door. A side-hinged rear door shall be provided, reinforced to prevent deformation which may cause leaks between the door and body. The door shall be hinged on the curbside with a minimum of three sets of heavy duty hinges. The door shall be held in the closed position by upper and lower heavy duty latches operated by a single lever type handle.

3.6.4.4 Load opening. A load opening shall be provided in the lower portion of the door through which the stationary compactor's packing ram may reciprocate. This opening shall be secured by a roll-down, chain-reinforced canvas cover, steel cover, or other adequate device(s). When a canvas cover is provided, suitable fastenings shall be provided for securing the cover in both the open and closed positions.

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3.6.4.4.1 Pin-off. Pin-off defines the procedure whereby compacted refuse is temporarily retained within the detachable body while the body is separated from the stationary compactor for loading onto the tilt-frame truck. When pin-off capability is specified (see 6.2), it shall be of the cable type. Provision shall be made on the detachable body to secure the pin-off cable to the body when the compactor ram is fully retracted, to retain compacted refuse within the detachable body until the load opening is secured.

3.6.4.5 Reflectors. The body shall have reflectors mounted in accordance with Federal Motor Carrier Safety Regulations 393.12 and 393.26(a) through (d).

3.6.4.6 Compatibility. When compatibility with an existing system is specified (see 3.6), and a retaining device (see 3.5.1.2.2) is provided on the existing tilt-frame truck, the body shall have means for securely engaging the retaining device. The detachable body shall mate with the existing stationary compactor and shall allow the compactor ram to reciprocate without binding or bypassing refuse. The body shall have a suitable means for engaging mating locking devices on the existing stationary compactor. Locking devices shall be securely fastened to the body to withstand the full compaction force of the compactor, including boost force. Construction of the body top, bottom and walls shall withstand the full compaction force of the compactor, including boost force, without permanent deformation.

3.6.5 Type G (stationary compactor). Type G stationary compactor shall be designed for use with the type F detachable body specified herein or when specified (see 3.6), with the detachable body specified. The stationary compactor shall consist of a frame structure, hopper, packer guide, reciprocating packing mechanism, and shield for the compaction of loose refuse in the detachable bodies. Compaction forces shall be accomplished by a hydraulically powered combination consisting of a double-acting cylinder, a power unit and controls. The stationary compactor frame and guides shall enable positive mating alinement with the charger opening of the detachable body. The stationary compactor shall be provided with not less than the cubic yard capacity, dimensions and force ratings specified in table V:

TABLE V. Compactor dimension and force rating requirements (minimum).

Size	NSWMA base size (cu. yds.)	Clear top opening (inches) Length	Width	Ram penetration (inches)	Force ratings (ram force in pounds) normal	maximum
1.5	1.5	42.0	58.0	16	35,300	39,200
4	4	78.0	58.0	20	75,600	90,000

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3.6.5.1 Construction. The stationary compactor shall be of heavy steel construction with a minimum of 1/4-inch plate for the face, sides, top shield and bottom. It shall be reinforced by a minimum of 6-inch structural channels and 4-inch structural channels along the sides, top shield and bottom. The stationary compactor shall be braced and supported by a heavy steel frame which shall be constructed to withstand heavy duty use. The stationary compactor shall be braced against reactionary forces of the ram cylinder. Foot pads and base members shall be provided for weight distribution and a means for securely anchoring the stationary compactor. The frame at the rear shall be an open structure in which a self-contained power unit shall be located for operation of the stationary compactor.

3.6.5.2 Ram mechanism. The ram mechanism and shield shall be powered by a double-acting hydraulic cylinder. The ram head shall transfer the refuse from the charging hopper of the compactor into the detachable bodies. Ram head guides shall be furnished to provide continuous sliding surface at the top and bottom of the compactor. A replaceable scraper shall be affixed along the back wall of the charging box to prevent refuse feed-back and to sweep the top plate as the ram mechanism is retracted. When pin-off capability is specified (see 3.6.4.4.1), means shall be provided to accomplish cable pin-off.

3.6.5.3 Ram cycle. The ram cycle shall be accomplished in not more than fifty-five seconds. A cycle shall be considered as a full outward travel and return of the ram mechanism.

3.6.5.4 Guide tracks. Guide tracks shall be provided and shall have a minimum length of 10 feet that shall extend forward of the stationary compactor and shall ensure positive mating of a detachable body and the stationary compactor. The driver shall be able to couple a detachable body to the stationary compactor without outside assistance and without leaving the cab of the truck.

3.6.5.5 Locking hooks. Locking hooks shall be located on the sides at the front of the stationary compactor for locking a detachable body in the mated position with the stationary compactor. The hooks shall automatically engage the lugs on each side of the detachable body when brought into mating position with the stationary compactor and shall be manually uncoupled from either side by a lever type handle. The locking hooks shall be constructed of high tensile steel having a minimum of 100,000 pounds per square inch yield strength.

3.6.5.6 Power unit. The power unit shall be self-contained, compact, and mounted within the stationary compactor frame at the rear end. The unit shall be provided with skids for easy removal. The power unit shall be provided with a hydraulic fluid reservoir tank of sufficient capacity to allow continuous 12 hour operation without overheating. The reservoir shall be furnished with one cleanout opening and an adequate filter. It shall be mounted to facilitate cleaning and servicing. An open, drip-proof, electric

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motor shall be furnished. The motor shall be of not less than 10 horsepower for size 1-1/2 compactor, and not less than 30 horsepower for size 4 compactors. Unless otherwise specified (see 6.2), the motor shall be for operation on 208 volts ac, 3-phase, 60 Hz power. Controls, bypass and check valves with interconnecting piping, fittings, and an oil tank shall be furnished.

3.6.5.7 Electric controls. The stationary compactor shall be furnished with a starter and disconnect operating on the voltage specified (see 3.6.5.6). The push button operation shall be controlled by 115-volt ac, 60 Hz power. All electrical components shall be mounted with NEMA 12 enclosures. The stationary compactor operation shall be by electrical controls through push buttons and shall be capable of being pre-programmed for automatic cycles.

3.6.5.8 Programmed cycle. Automatic continuous cycle, automatic one cycle, adjustable timed multiple stroke cycle or manual control shall be capable of being programmed by the user.

3.6.5.8.1 Automatic continuous cycle. When programmed for one cycle, one push button shall start the motor and pump, and also start the ram mechanism that shall reciprocate continuously until a push button shall stop the cycling.

3.6.5.8.2 Automatic one cycle. When programmed for one cycle, one push button shall start the motor and pump, and also shall start the ram mechanism for one complete cycle. Both the motor and the ram shall stop automatically.

3.6.5.8.3 Adjustable timed cycle. When programmed for timed, multiple-cycle, one push button shall start the motor and pump and also start the ram mechanism for the desired number of strokes (1 to 6 stroke selection). Both the motor and the pump shall stop automatically.

3.6.5.8.4 Manual control. When operating on manual control, push buttons shall be provided to: (a) start the motor and pump, (b) extend the ram mechanism, and (c) retract the ram mechanism. An emergency button shall be provided to stop the motor and pump. Boost pressure shall be obtained on manual control.

3.6.5.9 Push button panel. A push button panel shall be furnished. The panel shall be connected to the power unit by a neoprene covered multiconductor cable of the length specified (see 6.2). The panel shall consist of not less than:

(a) A key operated "off-on" switch. When a remote pendant is furnished, the switch shall be of the deadman type
(b) Manual-auto selector
(c) Start motor and pump button

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(d) Manual ram pack button
(e) Manual ram return button
(f) Start cycle button
(g) Stop cycle button
(h) Timed cycle selector (see 3.6.5.8.3)
(i) Boost "off-on" key switch.

3.6.5.10 Compatibility. When compatibility with an existing system is specified (see 3.6), the stationary compactor shall mate with the existing detachable bodies and allow the compactor ram to reciprocate without binding or bypassing refuse. The compactor shall have suitable means for engaging mating locking devices on the existing detachable bodies.

3.7 Lubrication. All moving parts of the equipment normally requiring lubrication shall have means provided for that lubrication. Pressure grease fittings shall be of the hydraulic type with surface check ball and when applied to sealed bearings shall have pressure release to protect the seals.

3.8 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: alinement of lights; adjustment of the engine, electrical and brake systems; filling and charging of the battery; alinement of the front wheels; inflation of all tires; complete lubrication of chassis, engine, and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives. Stationary compactors and detachable bodies shall be completely lubricated, as appropriate, with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.9 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle and equipment inspection. The first vehicle, each type of detachable body, and stationary compactor produced under the contract shall be inspected by the contractor under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle and equipment conformance to contractual requirements. Acceptance of the first production vehicle and equipment shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Road test. The vehicle, loaded to specified GVW (see 3.2.5 through 3.2.8), shall be driven for a distance of not less than ten miles over class I (concrete, asphalt, or brick) roads at maximum allowable traffic speed limits up to at least 55 mph. The load shall be discharged and the vehicle returned to the contractor's plant.

* 4.3.1.1 Weight. The first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on the front and rear axles. The total imposed loading on the front and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds and the payload required to provide the specified GVW. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

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4.3.2 Compatibility of hoist-haul unit with bodies. A detachable body of the make and model specified (see 6.2), shall be loaded and uncoupled from the tilt frame truck to ensure physical and functional compatibility. All operations of the unit and body shall be observed during pulling up of the detachable body, placing it in the carrying position, transporting it, dumping the contents with the body still mounted on the unit, and placing it back on the ground (loaded or empty) or, when specified (see 3.5), placing it on a dock. During these operations, the following shall be verified:

(a) the body control mechanism shall operate without malfunction
(b) the body shall mount properly in the carrying position
(c) the body lids shall open without interference or binding
(d) the body shall empty completely when the contents are dumped.

Upon completion of the operations, the tilt frame unit and body shall be examined for damage, misalinement, binding or other malfunction.

4.3.3 Detachable bodies. One of each type detachable body furnished under the contract shall be loaded and unloaded from a tilt frame truck of the make and model specified (see 6.2), to assure physical and functional compatibility.

4.3.4 Hoisting unit, type E body. The body hoisting mechanism on type E body shall be tested to ensure complete functional operation including the engagement, hoisting, and dumping of containers fully loaded with refuse, and the lowering and disengagement of the empty containers. Tests shall be made with the applicable class and each type and size of container specified in MIL-R-23954. Tests shall include random reversal of hoisting-lowering operations, operations from loading dock and ground level, and complete checkout of controls and safety system.

4.3.4.1 Containers for test. For the test, the contractor shall furnish standard containers of the applicable class of each type and size specified in MIL-R-23954.

4.3.5 Compaction unit, type C and E bodies. The compaction unit for type C and E bodies shall be tested using refuse to ensure complete functional operation of the compaction ram and for type E bodies to ensure complete ejection of the compacted refuse.

4.3.6 Type F body and type G stationary compactor. Type G stationary compactor shall be cycled without refuse to ensure complete functional operation of the ram and various cycles of operation (see 3.6.5.8). Test shall verify ram pressures conforming to the requirements of table V (see 3.6.5). The type F detachable body shall be coupled to and operated with a type G stationary compactor of the make and model specified (see 6.2) to ensure physical and functional compatibility. The type G stationary compactor shall be coupled and operated with a type F detachable body of the make and model specified (see 6.2), to ensure physical and functional compatibility.

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4.3.7 Examination. When testing is complete, the vehicle or detachable bodies shall be cleaned completely. The vehicle, bodies and compactors furnished for the test shall be examined for evidence of distortion, malfunction, and damage. The contractor shall be responsible for correcting defects disclosed during the operation and for taking corrective action to prevent recurrence of such defects in subsequently furnished vehicles.

4.3.8 Body treatment and painting. The certification regarding the cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to Government representatives at the first production vehicle inspection.

4.3.9 Inspection failure. Failure of the first production vehicle or equipment to meet requirements of the contract shall be cause for the Government to refuse acceptance of the vehicle or equipment until corrective action has been taken.

4.3.10 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, ensure that the vehicle, detachable body, and compactor conform to the physical and dimensional requirements and are capable of meeting the performance requirements specified herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government, indicating that the servicing and adjusting required by 3.8 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The truck, bodies, and stationary compactor covered by this specification are intended for nontactical use in the storage, transfer, and transport of refuse and other material generated at military installations.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Truck with tilt-frame, if required (see 1.1).

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(c) Type and size of detachable body, if required, or stationary compactor, if required (see 1.2 and 3.6).
* (d) Issues of DODISS to be cited in the solicitation and if required, the specific issues of individual documents referenced (see 2.1.1 and 2.2).
(e) The appropriate service for painting and marking (see 3.1.1.1).
(f) Rustproofing, if required (see 3.1.1.2).
(g) Tropical rustproofing, if required (see 3.1.1.2).
(h) Silicone rubber radiator and heater hoses, if required (see 3.4.1.7).
(i) Power plant heaters, if required (see 3.4.1.8).
(j) If battery heaters are not required with the power plant heaters (see 3.4.1.8).
(k) Dry type air cleaner, if required (see 3.4.3.1).
(l) A fuel and water separator, if required (see 3.4.3.3).
* (m) Oil lubricated wheel bearings and axle spindles, if required (see 3.4.9).
(n) Walking beam suspension with rubber load cushions, if required (see 3.4.9.1).
(o) Disc type wheels, if required (see 3.4.10).
(p) Bias ply tires, if required (see 3.4.10.1).
(q) Brake controls for use from a towing vehicle, if required (see 3.4.11.2).
(r) Increased braking capability, if required (see 3.4.11.3).
(s) Individual adjustable driver's seat, if required (see 3.4.12).
(t) Air operated horn, in addition, if required (see 3.4.20).
(u) Back-up alarm, if required (see 3.4.23).
(v) If the ability of tilt-frame unit to handle type E self-loading and compaction bodies is required (see 3.5).
(w) If the ability of the tilt-frame unit to unload on a dock is required (see 3.5).
(x) Body control by the hydraulic cylinder system, or the hydraulic operated winch and cable system, if required (see 3.5.1.2).
(y) Device for tandem loading, if required (see 3.5.1.2.1).
(z) Secondary mechanical retaining device, if required (see 3.5.1.2.2).
(aa) Set-down capability on loading dock, if required (see 3.5.3).
(ab) Hydraulic fluid conforming to MIL-H-83282, if required (see 3.5.4.1).
(ac) Existing equipment with which units must be compatible, if required (see 3.6).
(ad) Lower portion of body liquid-tight, if required for type A and B bodies (see 3.6.1.2).
(ae) Crane eyes, if required (see 3.6.1.4).
(af) A tarpaulin, if required (see 3.6.1.4).
(ag) Type C compaction body hydraulic system power unit voltage requirements, if not as specified (see 3.6.2.5).
(ah) Make of existing equipment and class of containers to be handled by the type E body, if required to supplement system (see 3.6.3.3).

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(ai) Provision for cable type pin-off, if required (see 3.6.4.4.1).
(aj) Stationary compactor voltage requirements, if not as specified (see 3.6.5.6).
(ak) Length of control cable for type G stationary compactors (see 3.6.5.9).
(al) Make and model of detachable body(s) the hoist-haul unit is to be tested with (see 4.3.1.2).
(am) Make and model of tilt-frame unit the detachable body(s) are to be tested with (see 4.3.2).
(an) Make and model of type G stationary packer the model F detachable body is to be tested with (see 4.3.5).
(ao) Make and model of type F body the stationary packer is to be tested with (see 4.3.5).

* 6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type. Unless other conditions are cited in the contract, computations shall be made for normal atmospheric pressure, normal ambient air temperature, and still dry air. The factors to be used in predicting truckability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude factor. 1.00
Table 3	- Rolling factor. 1.613
Table 4	- Area factor. 0.240
Table 5	- Velocity factor. 250.000
Table 6	- Altitude factor. 1.00
Table 7	- Chassis friction horsepower. Use applicable power unit GVW (to nearest, higher, tabulated 1000 pounds), and the engine rpm (to nearest 100 revolutions), which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade factor. 0.75
Table 8A	- Correction factor. Not required.
Table 9	- Road factor. 0.0

* 6.4 Subject term (key word) listing.

Body, detachable
Compactor unit
Container handling
Hoist-haul unit
Tilt-frame.

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